Examples :-

① A generator is coupled to a steam turbine. Calculate the speed of the turbine if the alternating frequency (rotational) of the generated sinusoid is 50Hz.

→ Assume that the generator is a two pole generator.

→ Solution :-

Little background :-

* For one complete cycle of voltage ~~va~~ waveform, the coil/conductor move past one pair of N & S poles.

∴ P = total no. of poles = 2p (p - no. of pairs of pole).

f = frequency of generated voltage Hz.

N - speed of rotor (or, turbine) in rev/min.

$$f = \frac{\text{No. of cycles}}{\text{seconds}} = \frac{\text{No of cycles}}{\text{revolutions}} \times \frac{\text{revolutions}}{\text{seconds}}$$

$$= \frac{P}{2} \times \frac{N}{60}$$

$$\boxed{f = \frac{PN}{60 \times 2}}$$

∴ speed of steam turbine $= N = \dfrac{120f}{P} = \dfrac{120 \times 50}{2}$

$$\boxed{N = 3000 \text{ rev/min}}$$

② A hydro-turbine speed is maintained at 150 rev/min. Find the no. of poles to generate 50Hz sinusoidal output voltage.

Solution :- $N = \dfrac{120f}{P}$, $P = \dfrac{120f}{N}$

$$P = \frac{\cancel{120 \times 50}}{\cancel{150}} \quad \frac{120 \times 50}{150} = 40 \text{ poles.}$$

(or, 20 pole pairs)

③ In an AC excited circuit, $v(t) = 200\sin(100t - 20^\circ)$ V

& $i(t) = 20\sin(100t + 40^\circ)$ A

find R, Z & X.

Solution :- $V_m = 200$ V, $I_m = 20$ A.

$$\bar{V} = \frac{200\text{ V}}{\sqrt{2}}\angle -20^\circ, \quad \bar{I} = \frac{20}{\sqrt{2}}\text{ A}\angle 40^\circ$$

$$\bar{Z} = \frac{\bar{V}}{\bar{I}} = \frac{200\angle -20^\circ}{20\angle 40^\circ} = 10\angle -20^\circ - 40^\circ = 10\angle -60^\circ\,\Omega$$

explantion - $\bar{Z} = \dfrac{200\angle -20^\circ}{20\angle 40^\circ} = \dfrac{200 \cdot e^{-j20}}{20\cdot e^{j40}}$

$$\bar{Z} = 10 \cdot e^{-j(20-40)} = 10e^{-j60}$$

$$\bar{Z} = 10\angle -60^\circ\,\Omega = 10(\cos 60 - j\sin 60)$$

$$\bar{Z} = 5 - 8.66j\,\Omega = R - jX_C$$

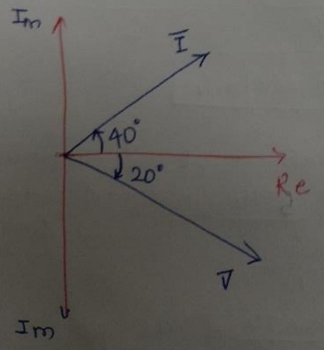
$$\therefore \boxed{R = 5\Omega}, \boxed{X_C = 8.66\,\Omega}$$

$$|Z| = \sqrt{5^2 + 8.66^2} = 10\Omega.$$

capacitor value canbe found from -

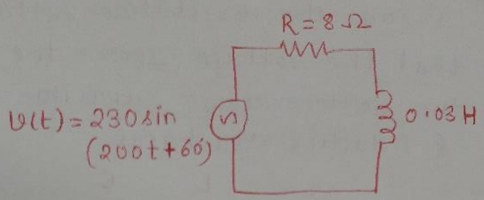
$$X_C = \frac{1}{\omega C}; \quad C = \frac{1}{X_C\omega} = \frac{1}{8.66 \times 100}$$

$$\boxed{C = 1.15\text{ mF}}$$



(* - Phasor diagram is drawn for illustration It is not needed to solve the problem).

④ For the given series R-L circuit, find the current.



Solution :- $X_L = j\omega L = j \times 200 \times 0.03 = 6j\ \Omega$.

$$Z = R + j\omega L = 8 + 6j\ \Omega = \sqrt{8^2+6^2}\ \angle \tan^{-1}\left(\frac{6}{8}\right)$$

$$Z = 10 \angle 36.87^\circ\ \Omega.$$

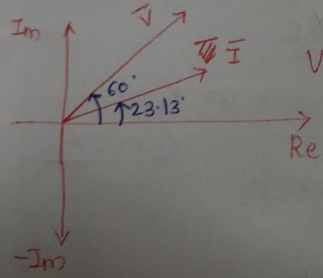
$$i(t) = \frac{v(t)}{Z} \quad \text{or} \quad \bar{I} = \frac{\bar{V}}{\bar{Z}}$$

$$= \cancel{230} \qquad = \frac{230 \angle 60^\circ}{10 \angle 36.87}$$

$$\bar{I} = 23 \angle 60 - 36.87 = 23.13^\circ$$

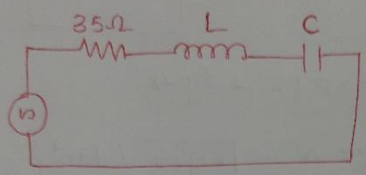
$$\therefore \bar{I} = 23 \angle 23.13^\circ\ A.$$

$$i(t) = 23 \sin(200t + 23.13^\circ)\ A.$$



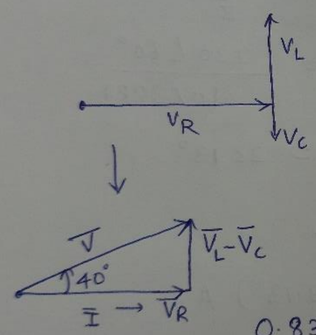
V leads I by 36.87°

⑤ For the following ckt, $R = 35\,\Omega$. The circuit current i lags the excitation voltage by $40°$. Assuming that the voltage across the ~~capacitor~~ inductor is 1.5 times the voltage across capacitor. find the capacitive & inductive reactance.



Solution :-

using phasor diagram approach -



$V_L = 1.5\,V_C \rightarrow$ (given).

$\hookrightarrow \therefore$ ckt is inductive in nature

$$\therefore \phi = \tan^{-1}\left(\frac{V_L - V_C}{V_R}\right)$$

$$40° = \tan^{-1}\left[\frac{X_L - X_C}{R}\right]$$

$$0.839\,R = X_L - X_C$$

$$0.839\,R = 0.5\,X_L \qquad \text{as } V_L = 1.5V_C,\ \underline{X_L = 1.5X_C}$$

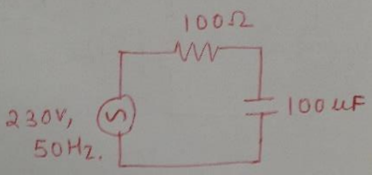
$$X_L = 1.678\,R = 58.73\,\Omega$$

$$X_C = \frac{X_L}{1.5} = 39.15\,\Omega.$$

Capacitive Reactance = $39.15\,\Omega$, Inductive Reactance = $58.73\,\Omega$

Note, Since the excitation frequency is not mentioned, the value of L & C can not be determined.

⑥ In the following series R-C circuit, $R = 100\Omega$, $C = 100\mu F$. Find – (i) current, i (ii) Phase shift between V & I (iii) V_C (iv) V_R (v) Power consumed.



Solution :-

$$\text{Impedance } Z = R - \frac{j}{\omega C}$$

$$Z = 100\Omega - \frac{j}{2\pi \times 50 \times 100 \times 10^{-6}}$$

$$Z = 100\Omega - 31.83j\ \Omega = 104.94\Omega \angle -17.656^\circ$$

(i) $$\bar{I} = \frac{\bar{V}}{\bar{Z}} = \frac{230\angle 0^\circ}{104.94\angle -17.656} = 2.192\text{ A} \angle +17.656^\circ$$

(ii) current leads voltage by $\underline{17.656^\circ}$.

instantaneous expression of voltage & current -

$v(t) = 230 \times \sqrt{2} \sin \omega t$ V. (230V - RMS value)

$i(t) = 2.192 \times \sqrt{2} \sin(\omega t + 17.656^\circ)$ A.

$$\boxed{\begin{aligned}\omega &= 2\pi f = 2\pi \times 50 \\ &= 100\pi = 314 \text{ rad/s}\end{aligned}}$$

Voltage across Cap.

(iii) $V_C = I \times X_C$

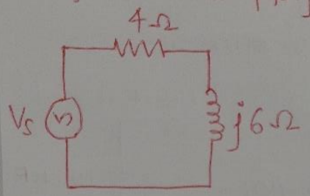
$= 2.192 \times 31.83 = \underline{69.77V.}$

(iv) $V_R = I \times R = 2.192 \times 100 = 219.2V.$

(v) Power consumed = Power dissipated in R = I^2R

$= 100 \times 2.192^2 = \underline{\underline{480.48 \text{ W}}}$

(7) For the below given R-L circuit, $I = 4A$. (RMS value). find the supply voltage.



* Solution :-

$$Z = 4 + j6 \ \Omega$$

$$|Z| = \sqrt{4^2+6^2} = 7.21\,\Omega$$

$$\phi = \tan^{-1}\left(\frac{6}{4}\right) = 56.31^\circ$$

$$\bar{V} = \bar{Z}\,\bar{I}$$

$$= 7.21 \angle 56.31^\circ \times 4A \angle 0^\circ$$

$$\bar{V} = 28.84 \angle 56.31^\circ$$

if, $i(t) = 4\sqrt{2}\sin\omega t$ A then -

$\hookrightarrow$ $v(t) = 28.84\sqrt{2} \times \sin(\omega t + 56.31^\circ)$

Voltage leads current by 56.31°.

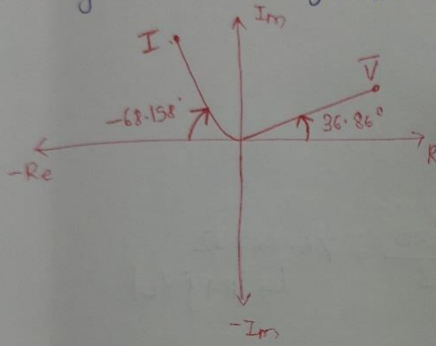
(5) An AC voltage $= (80+60j)V$ is applied to a circuit. The current flowing through the circuit is $(-4+10j)A$. Find impedance, power consumed & phase angle.

* Solution :-

$$V = 80+60j = 100\angle 36.86^\circ \ V$$

$$I = -4+10j = 10.77\angle -68.198 \ A.$$

using phasor diagram-



Net phase shift betn I & V

$$\phi = 180^\circ - [36.86^\circ + 68.198^\circ]$$

$$\boxed{\phi = 74.942^\circ}$$

- Current leads voltage by $\underline{74.942^\circ}$

power factor $= \cos\phi = 0.2597$ (leading).

power consumed $= VI\cos\phi$

$= 100 \times 10.77 \times \cos\phi$

$= 279.69 \approx \underline{280W}.$

Alternate way to compute power consumed.

Power consumed $= Re\{\bar{V}\bar{I}^*\}$, $\bar{I} = -4+10j$

$\bar{I}^* = -4-10j$

Power consumed $= Re\{(80+60j)\times(-4-10j)\}$

$= Re\{-320-800j-240j+600\} = Re\{\underline{280}-1040j\}$

$\therefore$ Power consumed $= \underline{\underline{280W}}$ [Note- -1040 is the reactive power]

Immersion (coil)

(9) A heater consumes 3A of current when powered from a 230V, 50Hz AC. The heater consumes 300W of power. Calculate resistance, impedance & inductance of the heater.

Note :-

Solution :-

Power consumed $= I^2R$

$300 = 3^2 \times R$

$\boxed{R = 33.33\,\Omega}$

$V = I \times Z$

$\therefore Z = \frac{V}{I} = \frac{230}{3} = 76.667\,\Omega$

$\hookrightarrow |R + jX_L|$

$Z^2 = R^2 + X_L^2$

$76.66^2 = 33.33^2 + X_L^2$

$X_L = \sqrt{76.66^2 - 33.33^2}$

$X_L = 69.035\,\Omega$

$\therefore \boxed{Z = 33.33 + j\,69.035\,\Omega}$

$X_L = \omega L = 69.035$

$\boxed{\therefore L = \frac{69.035}{2\pi \times 50} = 0.219\,H}$

(10) A
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a)

b)

c)

d)

(10) A coil of resistance 15Ω & inductance 60mH, a pure non-inductive resistor of 30Ω & loss-free 30 μF capacitor are connected in series across 230V, 50Hz supply.

Calculate: (a) Current

(b) Voltage across capacitor & the coil

(c) Power consumed in circuit

(d) Power consumed by coil

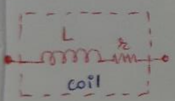
(e) Power factor of the circuit.

* Solution :- Total Resistance = Res. of coil + Resistor

$= 15 + 30 = \underline{45\Omega}.$

$X_L = 2\pi f L = 2\pi \times 50 \times 60 \times 10^{-3}$

$\underline{X_L = 18.85\,\Omega}$



$X_C = \dfrac{1}{2\pi f C} = \underline{106.10\,\Omega}$

Coil impedance $= |Z|_c = \sqrt{r^2 + X_L^2} = \sqrt{15^2 + 18.85^2}$

$\underline{|Z|_c = 24.09\,\Omega.}$

Net impedance of ckt $Z_T = R + j(X_L - X_C)$

$|Z|_T = \sqrt{R^2 + (X_L - X_C)^2} = \underline{98.17\,\Omega.}$

a) Current, $I = \dfrac{V}{|Z_T|} = \dfrac{230}{98.17} = 2.342\,A.$ (rms).

b) Voltage across cap $= I \times X_c = 248.48\,V,$

Voltage across coil $= I \times X_L = 44.1467\,V.$

c) Power consumed in ckt $= I^2 \times R_T = \cancel{105.39\,W}$ $I^2 \times 45 = \underline{\underline{246.82\,W}}$

d) Power consumed in coil $= I^2 \times r = 82.27\,W.$

e) Power factor, $\cos\phi = \dfrac{R}{Z_T} = \dfrac{45}{98.17} = 0.458$

* Since, $X_C > X_L$
pf is leading